AI-Driven Predictive and Prescriptive Analytics for Intelligent Decision Support Under Uncertainty

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***Abstract*—**This study develops and evaluates an AI-driven predictive-prescriptive analytics framework for intelligent decision support under uncertainty. The framework integrates multi-model regression, repeated cross-validation, statistical model comparison, conformal uncertainty quantification, model interpretation, and a prescriptive search layer. Experiments use the UCI Energy Efficiency benchmark with 768 simulated residential building configurations described by eight design variables and two targets: heating load and cooling load. A fixed split allocates 460 records to training, 154 to calibration, and 154 to final testing, while model robustness is additionally examined with 10-fold cross-validation repeated three times. Among Linear Regression, Random Forest, Gradient Boosting, a multilayer perceptron, and XGBoost, XGBoost achieved the strongest held-out performance (mean RMSE 0.6718; mean R^2 0.9942) and the lowest repeated-CV RMSE (0.6717 ± 0.0788). Split-conformal intervals produced empirical coverage of 85.1/81.8%, 93.5/93.5%, and 97.4/100% for heating/cooling load at nominal 80%, 90%, and 95% levels. The prescriptive layer, which searches feasible orientation and glazing alternatives, outperformed the random-mean and rule-based baselines in all test cases and was the overall winner against all evaluated baselines in 92.2% of cases. Mean model-estimated load reduction relative to the current configuration was 8.28%. The results show how accurate prediction, explicit uncertainty, and prescriptive recommendation can be combined into a transparent decision-support pipeline rather than treated as isolated analytics tasks.

***Index Terms*—**predictive analytics, prescriptive analytics, decision support, uncertainty quantification, conformal prediction, XGBoost, building energy, data-driven optimization.

I. INTRODUCTION

Data-driven decision support is most useful when it connects three questions that are often addressed separately: what is likely to happen, how uncertain is that prediction, and what action should be taken next. Predictive machine-learning models answer the first question, but point predictions alone can create false confidence when uncertainty is not exposed to the decision maker. Prescriptive analytics extends prediction toward action by evaluating alternative decisions and recommending options expected to improve an objective [8]. In high-consequence settings, the combination of predictive accuracy, uncertainty awareness, and actionability is more informative than any one component in isolation.

Building-energy analysis provides a compact but meaningful test bed for this integration. Data-driven methods have become established tools for estimating building energy demand and supporting design and operational decisions [1]-[3]. The Energy Efficiency benchmark introduced by Tsanas and Xifara contains 768 residential building configurations, eight design attributes, and simulated heating and cooling loads [1]. The feature space includes relative compactness, surface and wall area, roof area, overall height, orientation, glazing area, and glazing distribution. Because these variables include both structural characteristics and design choices, the dataset supports not only regression but also a limited counterfactual decision exercise.

This paper proposes an end-to-end framework that moves from prediction to uncertainty-aware prescription. Five regression models are compared on a common split, the ranking is tested under repeated cross-validation, and the leading model is subjected to paired significance tests. Split conformal prediction is used to attach distribution-free marginal prediction intervals to heating and cooling estimates [7]. A permutation-based interpretation layer identifies influential design attributes. Finally, a prescriptive search evaluates feasible alternatives for orientation, glazing area, and glazing distribution while holding structural geometry fixed, and compares the selected option against random and rule-based baselines. The resulting research question is: How effectively can AI-driven predictive analytics, uncertainty quantification, and prescriptive search be integrated to support transparent and data-driven decisions under uncertainty?

The contributions are fourfold: (1) a reproducible comparison of linear, tree-ensemble, boosting, and neural-network regressors; (2) repeated 10-fold cross-validation with statistical comparison of the leading model; (3) multi-level conformal uncertainty analysis at 80%, 90%, and 95% nominal coverage; and (4) a prescriptive decision layer validated against current, random-alternative, and fixed rule-based baselines. Importantly, prescriptive improvements are reported as model-estimated changes rather than measured physical energy savings.

II. RELATED WORK

A. Data-Driven Building-Energy Prediction

Tsanas and Xifara demonstrated that statistical machine-learning methods can estimate heating and cooling loads accurately from a small set of building-design variables [1]. Broader reviews show that data-driven building-energy research spans statistical regression, tree ensembles, neural networks, and hybrid approaches, with model performance depending strongly on feature design, target definition, data quality, and evaluation protocol [2], [3]. These studies motivate the use of multiple model families rather than assuming a single learning paradigm is universally best.

B. Ensemble Learning and Predictive Modeling

Random Forest combines randomized decision trees to reduce variance and provide robust nonlinear regression [5]. Gradient boosting instead constructs additive learners stage-by-stage to reduce predictive loss [6]. XGBoost extends gradient-boosted trees with regularization and systems optimizations for scalable learning [4]. These methods are well suited to heterogeneous tabular data and provide strong baselines for the energy-performance task considered here.

C. Uncertainty and Prescriptive Analytics

Prediction intervals are essential when a point estimate may be used to trigger a decision. Conformal methods provide finite-sample, distribution-free coverage guarantees under exchangeability, making them attractive when a full probabilistic model is unavailable [7]. Prescriptive analytics shifts the objective from predicting outcomes to choosing actions that improve a downstream criterion [8]. This study combines these ideas in a single pipeline: regression produces expected loads, conformal calibration quantifies uncertainty, and a constrained search recommends design alternatives.

III. DATA AND EXPERIMENTAL DESIGN

A. Dataset and Split

The experiment uses the Energy Efficiency dataset with 768 observations and eight predictors. The two continuous targets are heating load and cooling load. Using random seed 42, the

implementation creates 460 training records, 154 calibration records, and 154 held-out test records. The calibration partition is reserved for conformal residual estimation so that prediction intervals are not calibrated on the final test set.

TABLE I
DATASET VARIABLES

Variable	Role
Relative Compactness	Input
Surface Area	Input
Wall Area	Input
Roof Area	Input
Overall Height	Input
Orientation	Input
Glazing Area	Input
Glazing Area Distribution	Input
Heating Load	Target
Cooling Load	Target

B. Predictive Models and Metrics

Five models are evaluated: Linear Regression, Random Forest, Gradient Boosting, an MLP neural network, and XGBoost. Model quality is summarized by mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2), computed separately for the two targets and averaged for model ranking. Training and inference times are retained as secondary computational indicators. XGBoost is selected for subsequent uncertainty and prescriptive analysis because it has the lowest mean RMSE and highest mean R^2 on the held-out test set.

C. Repeated Cross-Validation and Statistical Tests

Robustness is evaluated using 10-fold cross-validation repeated three times, yielding 30 paired fold scores per model. Mean and standard deviation are reported for MAE, RMSE, and R^2. The XGBoost RMSE scores are compared with each alternative using a paired t-test and Wilcoxon signed-rank test. Because repeated cross-validation folds are statistically dependent, these tests are treated as supportive evidence rather than exact independent-sample inference; this limitation is consistent with prior cautions regarding significance testing across resampled learning experiments [9], [10].

D. Conformal Uncertainty Quantification

For each target, absolute residuals on the calibration partition are used to estimate split-conformal half-widths. Intervals are evaluated at nominal coverage levels of 80%, 90%, and 95%. The analysis reports empirical test coverage and mean interval width. The central decision trade-off is reliability versus informativeness: higher nominal coverage should generally increase empirical coverage, but also widen the interval presented to the decision maker.

E. Prescriptive Decision Layer

The prescriptive component changes only attributes that represent configurable design choices: orientation, glazing area, and glazing distribution. Structural geometry is held fixed. Candidate triples are restricted to combinations observed in the dataset with nonzero glazing, reducing extrapolation into implausible configurations. For each test case, XGBoost predicts heating and cooling loads for the feasible candidates. The objective minimizes predicted total load plus a small penalty for changing glazing area. The selected recommendation is compared with (i) the current design, (ii) the mean of 20 random valid alternatives, (iii) the best of the 20 random alternatives, and (iv) a fixed rule-based moderate-glazing configuration. This is a model-based counterfactual analysis; no recommended design was physically constructed or independently simulated.

IV. RESULTS

A. Held-Out Predictive Performance

TABLE II
HELD-OUT TEST PERFORMANCE

Model	MAE	RMSE	R^2
XGBoost	0.4784	0.6718	0.9942
Gradient Boosting	0.6339	0.9179	0.9887
Random Forest	0.7773	1.1865	0.9792
MLP	2.1460	2.9493	0.9075
Linear Regression	2.2318	3.0914	0.8987

XGBoost produced the strongest overall held-out results, with mean MAE 0.4784, RMSE 0.6718, and R^2 0.9942. The target-specific R^2 values were 0.9986 for heating load and 0.9898 for cooling load. Gradient Boosting ranked second, while Random Forest remained strong but less accurate. The MLP and linear model were substantially weaker, indicating that nonlinear tree ensembles capture the structure of this tabular design space more effectively.

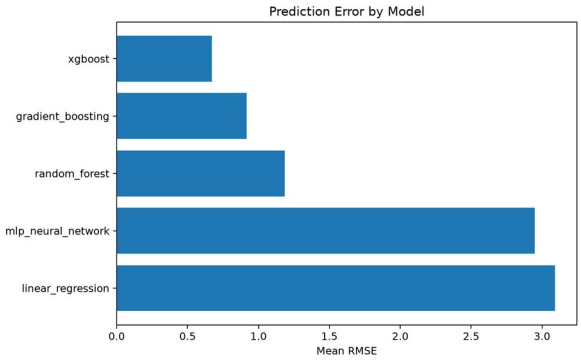


Fig. 1. Held-out RMSE comparison across the five predictive models.

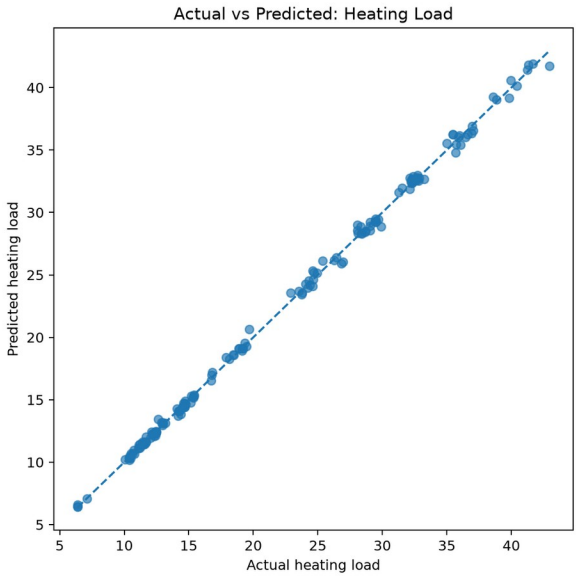


Fig. 2. Actual versus predicted heating load for the selected XGBoost model.

B. Repeated Cross-Validation

TABLE III
REPEATED 10-FOLD CROSS-VALIDATION (3 REPEATS)

Model	RMSE mean ± SD	R^2 mean ± SD
XGBoost	0.6717 ± 0.0788	0.9938 ± 0.0015
Gradient Boost.	0.9369 ± 0.0853	0.9875 ± 0.0023
Random Forest	1.1348 ± 0.1352	0.9803 ± 0.0052

MLP	2.9946 ± 0.3854	0.9023 ± 0.0236
Linear Reg.	3.0595 ± 0.3000	0.8992 ± 0.0180

The cross-validation ranking matches the held-out ranking. XGBoost achieved the lowest mean RMSE (0.6717) and the highest mean R^2 (0.9938), with relatively small fold-to-fold variation. Relative to Gradient Boosting, the nearest competitor, the mean RMSE advantage was approximately 0.265. The paired t-test gave $p = 5.36\times10^{-22}$ and the Wilcoxon test $p = 1.86\times10^{-9}$. Comparisons with Random Forest, MLP, and Linear Regression were also significant at the 0.05 level in both stored tests. These values support the empirical ranking, although the dependence among repeated cross-validation folds means the p-values should not be interpreted as if 30 independent experiments had been conducted.

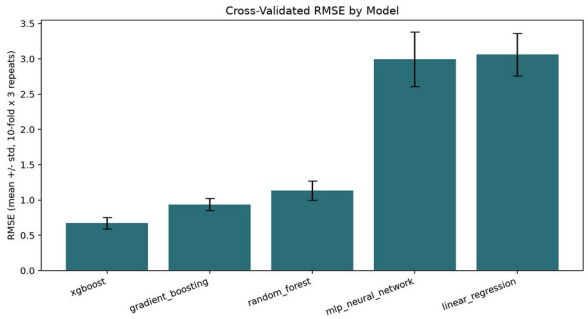


Fig. 3. Repeated cross-validation RMSE (10 folds × 3 repeats).

C. Uncertainty Coverage

TABLE IV
CONFORMAL COVERAGE AND INTERVAL WIDTH

Target	Nom.	Emp.	Width
Heating	80%	85.1%	1.035
Cooling	80%	81.8%	2.362
Heating	90%	93.5%	1.512
Cooling	90%	93.5%	3.837
Heating	95%	97.4%	1.883
Cooling	95%	100%	7.798

Empirical coverage was at or above the nominal level for every reported target-level pair in this held-out sample. The 90% intervals covered 93.5% of both heating and cooling observations. The expected coverage-width trade-off is clear: heating interval width rose from 1.035 at 80% nominal coverage to 1.883 at 95%, while cooling width rose more sharply from 2.362 to 7.798. Cooling-load uncertainty is therefore materially larger and should be communicated to decision makers rather than hidden behind a point prediction.

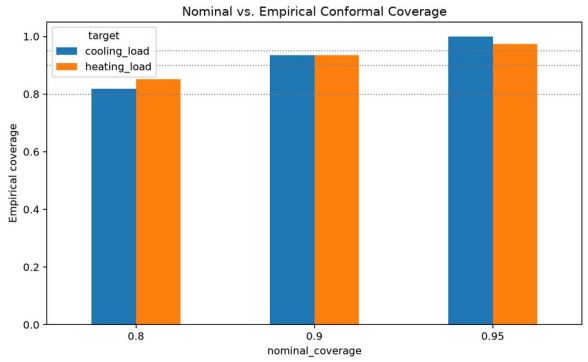


Fig. 4. Nominal versus empirical conformal coverage across uncertainty levels.

D. Explainability

Permutation importance identified relative compactness as the dominant predictor, with a mean importance of 1.3151 under the implemented scoring procedure. Glazing area ranked second (0.1121), followed by surface area (0.0287), glazing distribution (0.0168), orientation (0.0162), and wall area (0.0050). The ranking is consistent with the idea that building form and glazing choices strongly influence predicted thermal loads, but permutation importance measures predictive dependence in this fitted dataset and is not a causal estimate.

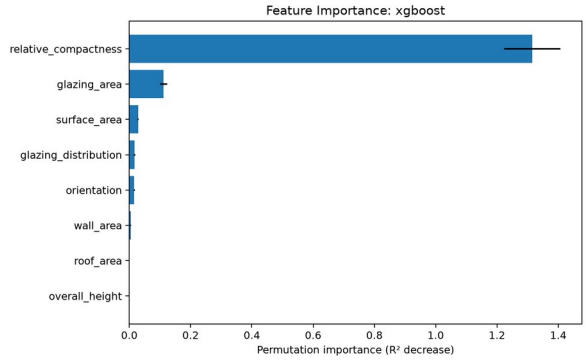


Fig. 5. Permutation feature importance for the selected XGBoost model.

E. Prescriptive Validation

TABLE V
PRESCRIPTIVE-LAYER VALIDATION

Measure	Result
Test designs	154
Mean reduction vs. current	4.598 load units
Mean reduction vs. current	8.28%
Beats random-mean baseline	100%
Beats rule-based baseline	100%
Overall winner vs. all baselines	92.21%

Across 154 test designs, the prescriptive search produced a mean model-estimated reduction of 4.598 total-load units relative to the current configuration, equivalent to 8.28% on average. It beat the random-mean and rule-based baselines in all cases and was the overall winner against all evaluated baselines in 92.21% of cases. The result supports the value of using the predictive model as a decision engine rather than only as a forecaster. However, some individual recommendations had negative estimated reduction relative to the current configuration, and the method does not represent construction cost, daylighting quality, code requirements beyond the simple candidate restrictions, or occupant preferences.

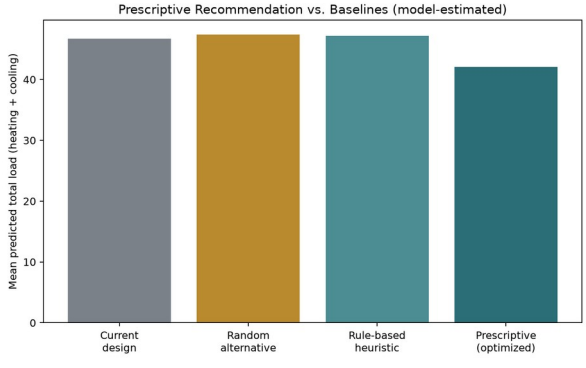


Fig. 6. Prescriptive recommendations compared with evaluated baselines.

V. DISCUSSION

The results support three conclusions. First, the predictive stage is

highly accurate on this benchmark and stable under resampling. XGBoost consistently outperformed the alternative models in both the fixed held-out split and repeated cross-validation. This is consistent with the effectiveness of gradient-boosted trees on structured tabular problems [4], [6]. The gain is not merely a single-split artifact because the ordering persists over 30 cross-validation folds.

Second, explicit uncertainty changes how predictions should be consumed. A point prediction of cooling load may appear precise, yet the conformal results show that the interval required for 95% empirical reliability can be wide. This distinction is central to intelligent decision support under uncertainty: confidence should influence whether a decision is automated, escalated for review, or treated as exploratory. The model can therefore supply not only an expected outcome but also a reliability envelope around that outcome.

Third, the prescriptive layer demonstrates a practical transition from prediction to action. The same XGBoost model that predicts heating and cooling loads is queried over feasible alternative configurations and used to rank actions. This is conceptually aligned with predictive-to-prescriptive analytics, in which auxiliary data and predictive models are embedded into a downstream decision problem [8]. The strong performance against random and fixed-rule baselines indicates that individualized recommendations are more effective within the modeled design space than non-adaptive alternatives.

The study also highlights an important distinction between decision intelligence and unconstrained optimization. Recommendations should not be interpreted as universally optimal building designs. The search is deliberately narrow, changing only orientation and glazing variables and using predicted thermal load as the central objective. A production decision-support system would require multi-objective criteria that may include capital cost, comfort, daylighting, carbon impact, feasibility, regulation, and stakeholder preference. The present framework is therefore best viewed as a validated analytical core that can be extended with domain-specific constraints and utility functions.

VI. LIMITATIONS AND THREATS TO VALIDITY

The dataset contains simulated building configurations rather than measurements from occupied buildings; consequently, external validity to real operational conditions, climate variability, control behavior, and occupant effects is limited. The sample is small (768 records), and the decision variables take a finite set of values. The conformal intervals provide marginal coverage under exchangeability but do not guarantee equal conditional coverage for every region of the feature space. The paired significance tests operate on dependent repeated-cross-validation folds, so their p-values are supportive rather than definitive. More rigorous corrected resampled tests or nested evaluation could be used in future work [9], [10].

The prescriptive analysis is also model-based. The reported 8.28% average improvement is a predicted counterfactual reduction in the model objective, not measured physical energy savings. Although candidate design triples are restricted to combinations observed in the dataset and the zero-glazing case is excluded, recommendations are not independently verified using a separate building-energy simulator. Finally, the objective function does not model financial cost, implementation constraints, uncertainty-aware robust optimization, or competing design goals.

VII. CONCLUSION AND FUTURE WORK

This paper presented an integrated AI-driven framework for predictive and prescriptive analytics under uncertainty. Using the Energy Efficiency benchmark, XGBoost achieved mean held-out RMSE 0.6718 and R^2 0.9942 and retained the strongest ranking under repeated 10-fold cross-validation. Conformal prediction transformed point estimates into calibrated uncertainty intervals whose empirical coverage tracked or exceeded nominal levels in the held-out sample. Permutation analysis identified relative compactness as the dominant predictive feature. The prescriptive search converted predictions into individualized design recommendations and outperformed random-mean and fixed rule-based baselines in all test cases, while winning against all evaluated baselines in 92.21% of cases. The resulting pipeline demonstrates that intelligent decision support can be organized as a sequence of prediction, validation, uncertainty quantification, interpretation, and constrained recommendation.

Future work should validate the prescriptive recommendations with an independent physical simulator or real building data, incorporate uncertainty directly into the optimization objective, use nested or corrected statistical evaluation, and extend the decision layer to multi-objective trade-offs involving cost, carbon, comfort, and feasibility. Additional work should study conditional conformal methods and more recent energy datasets to assess whether the framework remains reliable under distribution shift. The central objective is not simply to maximize predictive R^2, but to create decision support that communicates what is expected, how uncertain the expectation is, and which feasible action is preferred under explicit assumptions.

REPRODUCIBILITY STATEMENT

The accompanying project contains the dataset, executable Python scripts, model definitions, random seed (42), held-out predictions, repeated cross-validation fold results, significance-test output, conformal-coverage tables, feature importance, prescriptive-validation files, and all figures used in this manuscript. The primary experiment uses 460/154/154 training/calibration/test records, and the rigor experiment uses repeated 10-fold cross-validation with three repeats. Reported numerical results are taken directly from the saved project outputs.

REFERENCES

[1] A. Tsanas and A. Xifara, “Accurate quantitative estimation of energy performance of residential buildings using statistical machine learning tools,” Energy and Buildings, vol. 49, pp. 560–567, 2012, doi: 10.1016/j.enbuild.2012.03.003.

[2] K. Amasyali and N. M. El-Gohary, “A review of data-driven building energy consumption prediction studies,” Renewable and Sustainable Energy Reviews, vol. 81, pp. 1192–1205, 2018, doi: 10.1016/j.rser.2017.04.095.

[3] Y. Sun, F. Haghighat, and B. C. M. Fung, “A review of the-state-of-the-art in data-driven approaches for building energy prediction,” Energy and Buildings, vol. 221, Art. no. 110022, 2020, doi: 10.1016/j.enbuild.2020.110022.

[4] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[5] L. Breiman, “Random forests,” Machine Learning, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[6] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” The Annals of Statistics, vol. 29, no. 5, pp. 1189–1232, 2001, doi: 10.1214/aos/1013203451.

[7] J. Lei, M. G’Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, “Distribution-free predictive inference for regression,” Journal of the American Statistical Association, vol. 113, no. 523, pp. 1094–1111, 2018, doi: 10.1080/01621459.2017.1307116.

[8] D. Bertsimas and N. Kallus, “From predictive to prescriptive analytics,” Management Science, vol. 66, no. 3, pp. 1025–1044, 2020, doi: 10.1287/mnsc.2018.3253.

[9] T. G. Dietterich, “Approximate statistical tests for comparing supervised classification learning algorithms,” Neural Computation, vol. 10, no. 7, pp. 1895–1923, 1998, doi: 10.1162/089976698300017197.

[10] C. Nadeau and Y. Bengio, “Inference for the generalization error,” Machine Learning, vol. 52, no. 3, pp. 239–281, 2003, doi: 10.1023/A:1024068626366.